The Informationally Integrated Alignment Hypothesis: Integrated Information Aligns with and Predicts Reward in Reinforcement Learning Agents

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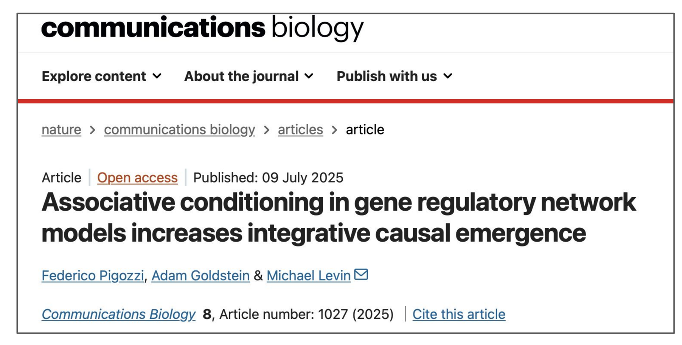
INTRODUCTION

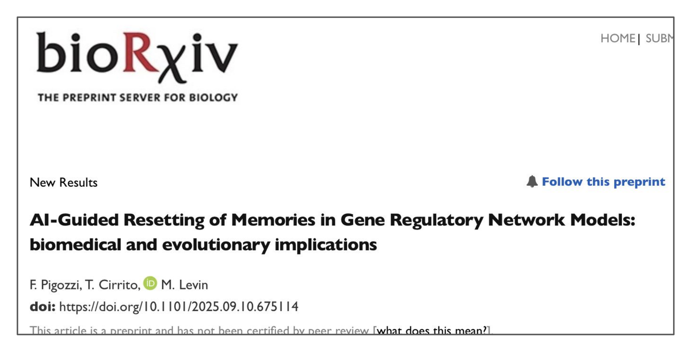
- Agents on Earth exert causal power and are drivers of subsequent events
- There's math (**integrated Information**) that measures an agent's predictive power on its future

We have shown that:

- Biological agents increase their integrated information **after learning**

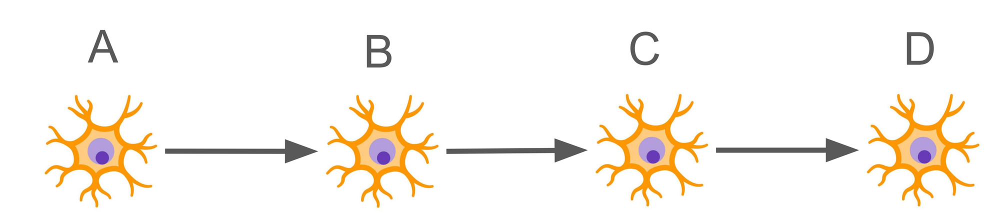
So what about learning in artificial agents?



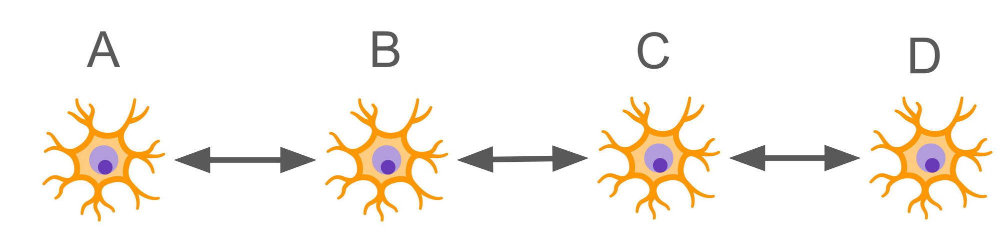


INTEGRATED INFORMATION ϕ

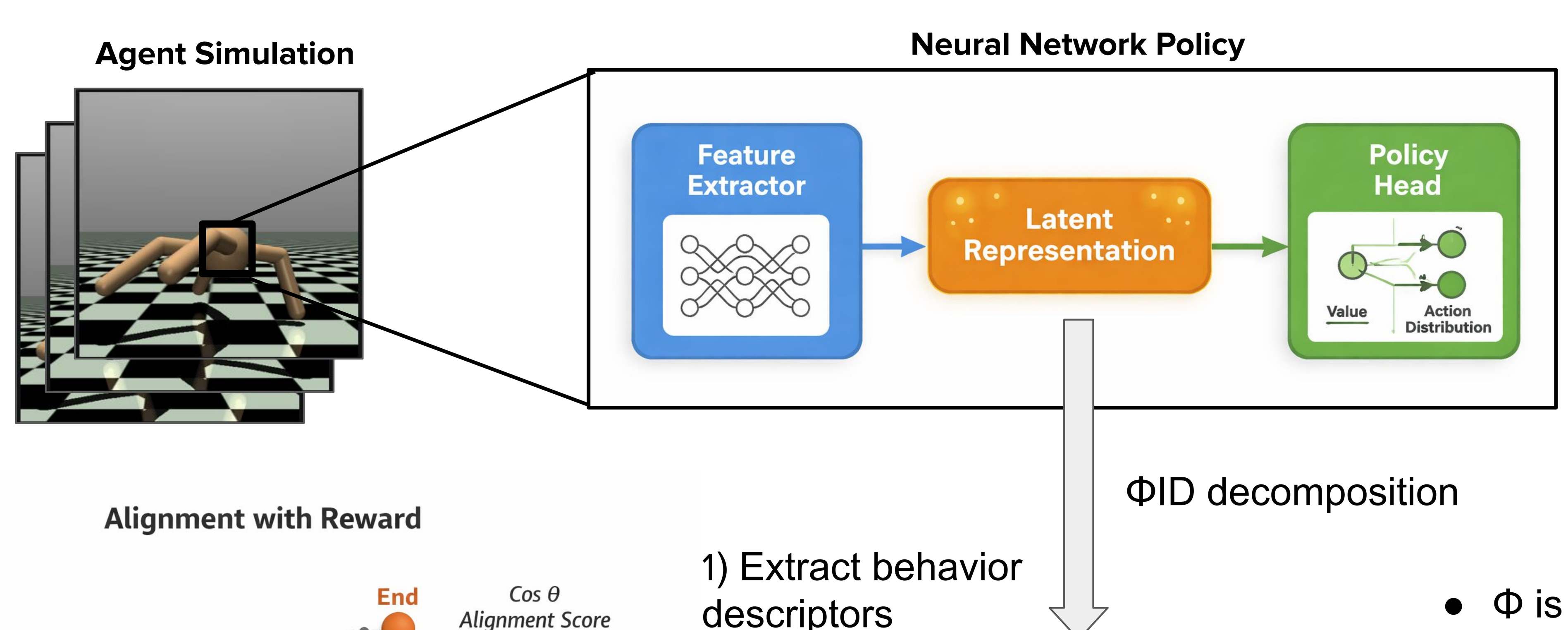
- How much is the system a **whole**, rather than sum of **parts**?
- Low Φ:

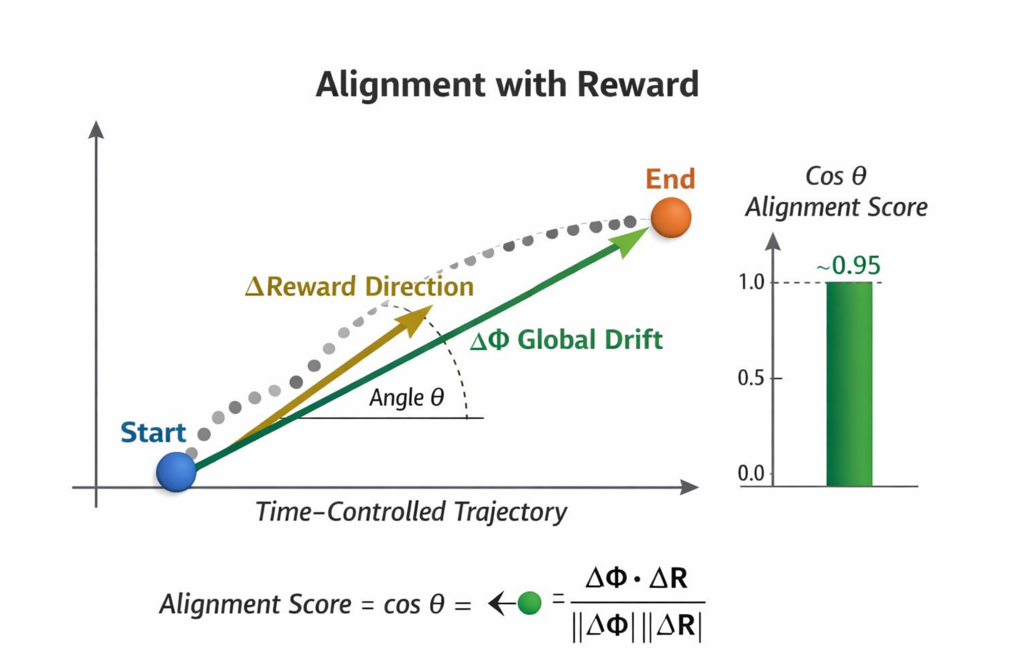


- High Φ:

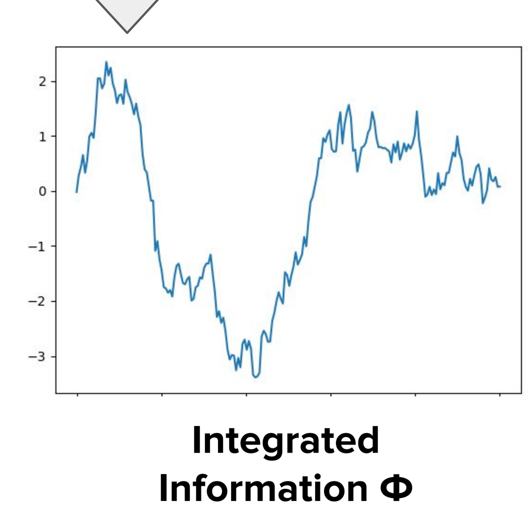


METHODS





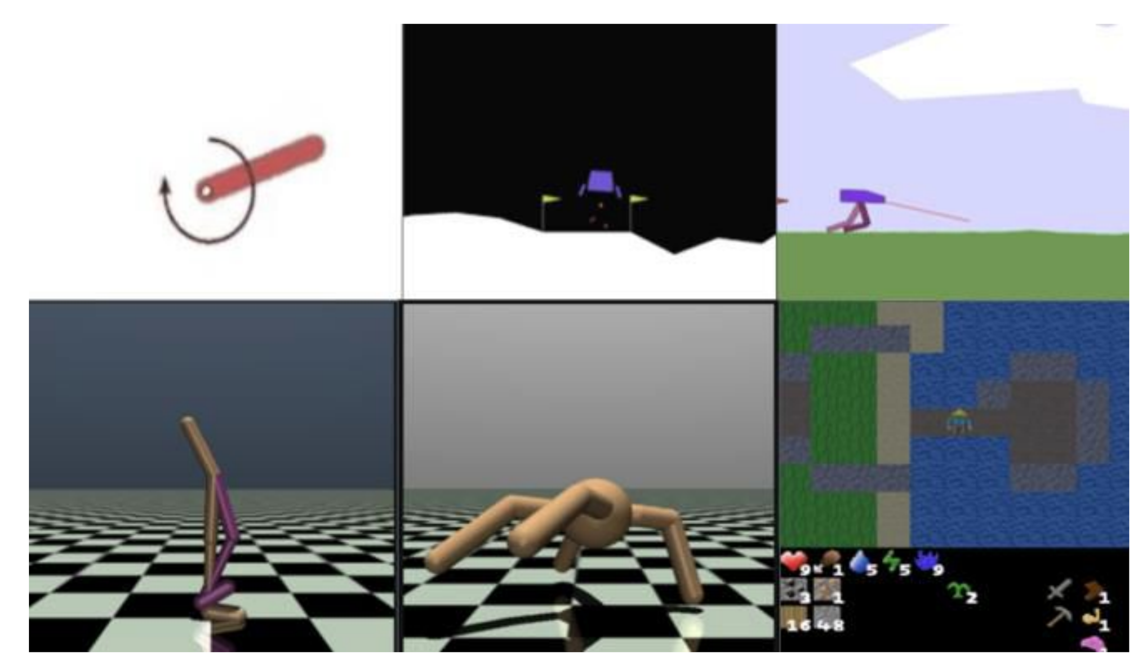
1) Extract behavior descriptors
2) Embed in 2D with PCA



- Φ is a trajectory over sim time.
- It measures how much information the whole provides about the future evolution cannot be inferred by any subset of parts.

EXPERIMENTS

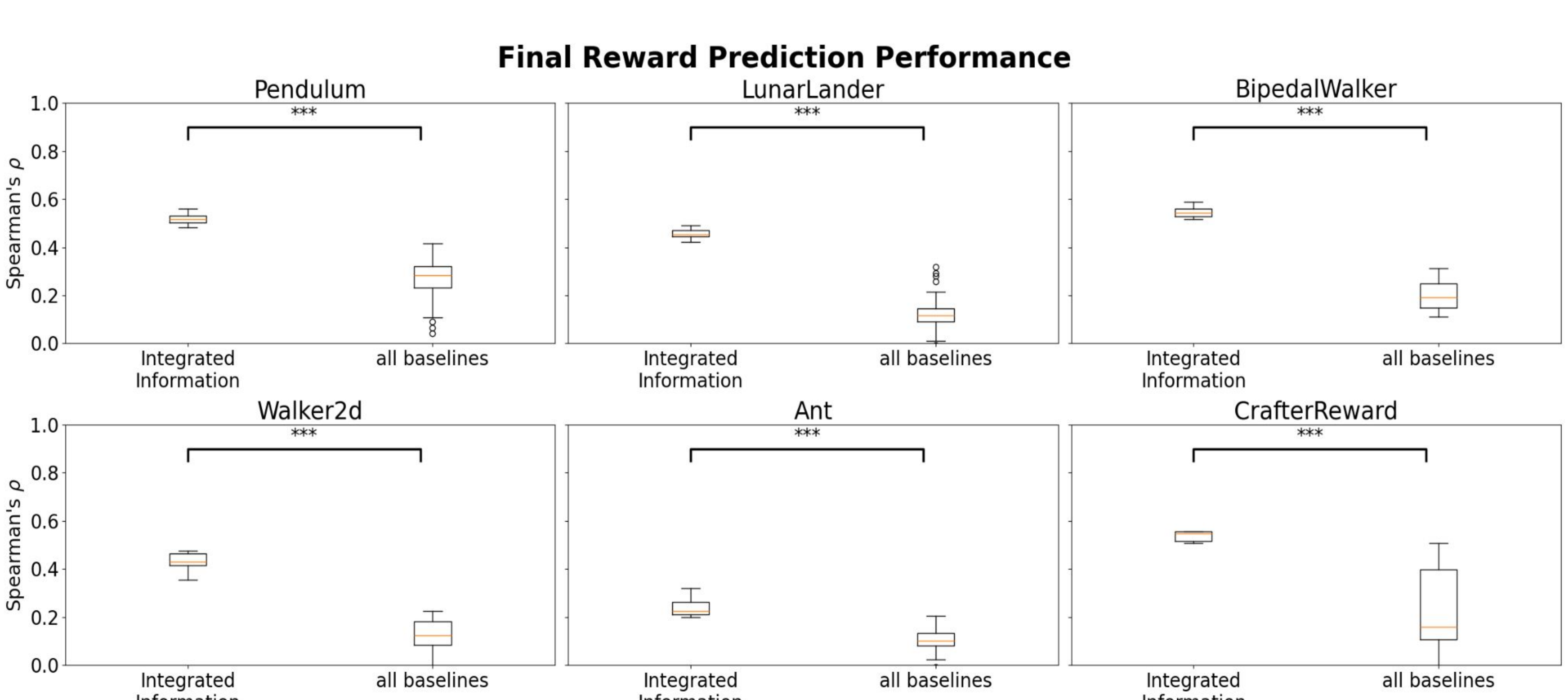
- 10 random seeds.
- 2 policy architectures (MLP and GRU).
- 2 algorithms (PPO and SAC/DQN).
- 6 tasks on complexity spectrum:



RESULTS

	Global Reward Alignment	Local Reward Alignment
Pendulum	0.99	0.00
LunarLander	1.0	0.00
BipedalWalker	0.86	0.00
Walker2D	0.35	0.03
Ant	0.49	0.02
CrafterReward	-0.95	0.00

- **Reward alignment**: the degree to which Φ in embedding space aligned with the direction that increased reward.
- **Global** for the whole latent trajectory.
- **Local** as the mean of the instantaneous angles.
- Global alignment was **strong** in magnitude, while the local alignment was **negligible**.
- Slow **representational drift** relating to learning.



- Random forest trained to **predict final reward**, given the measure on the x-axis: Φ descriptors or all the baseline representational metrics..
- Φ significantly **outperformed** in all environments.
- Φ works like a **compression** of existing representational metrics.

CONCLUSIONS

- Successful agents have Φ representations that **align** with reward and
- Are **predictive** of final learning outcomes.
- An agent's border is determined by the size of its goals.
- As cells adapt to reach **homeostasis**, RL agents reach a preferred state and establish selves.

LIMITATIONS

- Alignment and prediction do **not** imply **causation.**
- Missing a **mechanistic** interpretation.
- **Intervening on phi** to drive reward up (in the works!)

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Paper